

BobaLink Browser Extension “ Master Implementation Plan

Revision: 1 **Last modified:** 2026-06-10T05:12:40Z **Status:** Planning complete “ ready for phased, subagent-driven implementation **Authority:** This plan is derived exhaustively from docs/research/browser_extension/ (205 files, ~50k lines) via the analysis files in _analysis/01“06.md and the planning artifacts in _plan/A“F.md. Per “11.4.118, the traceability matrix (_plan/C-traceability-matrix.md, 245 items) proves nothing is skipped.

0. Hard constraints (non-negotiable “ read first)

1. **NO CI/CD “ EVER.** No .github/workflows/, no .gitlab-ci.yml, no Jenkins, no pipelines, no git hooks added for this subproject. The reference ships ci.yml/release.yml “ they are **DROPPED**; their build/test/lint *commands* are extracted into a manual extension/ci-ext.sh shelled from the repo“™s ./ci.sh. (Operator-reinforced 2026-06-10; Boba Hard Stop.)
 2. **Anti-bluff (“11.4 / “11.4.107).** Every test asserts a user-observable outcome and must fail against a no-op stub. The reference suite“™s 10 known bluffs (_plan/D) are remediated, not adopted.
 3. “11.4.10 **credentials.** The extension stores **no decryptable secret by default** (delegate-by-default, _plan/E). The hard-coded "boba-link-extension" passphrase + empty-passphrase decrypt are **forbidden**. No real credential in any test/fixture/doc.
 4. “11.4.113 **no force-push**; “11.4.71 **fetch-first**; **SSH remotes only**.
 5. **100% multi-type test coverage** (“11.4.27/“11.4.85): unit, integration, e2e, security, load, scaling, chaos, stress, performance, benchmark, UI, UX, Challenges, HelixQA banks + autonomous QA sessions “ for every feature/flow/use-case/edge-case.
 6. “11.4.125/“11.4.134/“11.4.142 **mandatory code-review** of every change before it is “œcomplete,“œ iterate-until-GO.
 7. **Real backend = Boba merge-service on 7187** (_plan/A), not direct qBittorrent, not port 8080.
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1. Product definition

BobaLink “ a cross-browser **Manifest V3 WebExtension** (WXT + TypeScript strict, Vitest+Playwright) that: - detects **magnet links** + **.torrent files** on any page (MutationObserver, SPA-aware) and across **tab groups**; - parses them (bencode / magnet BEP-9 / .torrent “ SHA-1 infohash), dedups by infohash; - forwards them to **Boba“™s merge-service** (: 7187), which adds them to qBittorrent server-side; - provides popup + options UI, offline FIFO queue, badge/notifications, context menu, keyboard shortcuts, i18n (8 locales), WCAG 2.1 AA, dark/light themes; - targets Chrome “¥109, Firefox “¥109 (MV3 event pages via WXT), Opera, Edge, Yandex (Safari deferred).

Supported browsers / data model / sequences / state machines: see _analysis/03-diagrams.md (6 ER entities, 2 end-to-end sequences, offline-queue + lifecycle state machines, security architecture, compatibility matrix).

2. Repository placement & naming (proposed “operator-objectable”)

- **Location:** new top-level extension/ directory (sibling to frontend/, download-proxy/). Snake_case per §11.4.29 where it doesn’t break tooling; the WXT/TS conventions inside extension/src/ follow the framework.
 - **Name:** BobaLink (from the materials). Objectable.
 - **Build artifact:** extension/.output/chrome-mv3-prod/ (+ firefox), zipped for manual store submission (no auto-submit).
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3. Backend additions required in Boba (Python merge-service)

From `_plan/A` the extension needs capabilities Boba doesn’t yet expose. These are **Phase 4** tasks (real, tested, anti-bluff): - **BE-1 “CORS for extension origins:** add `chrome-extension://* + moz-extension://*` to `ALLOWED_ORIGINS` (or document the host-permission background-fetch path that avoids CORS). File: `download-proxy/src/api/__init__.py` CORS middleware. - **BE-2 “raw .torrent upload:** new `POST /api/v1/download/upload` (multipart bytes → qBittorrent `/api/v2/torrents/add` multipart). Boba currently accepts URLs only. File: `download-proxy/src/api/routes.py`. - **BE-3 “Go-profile parity (optional):** qBittorrent-go/`internal/api/download.go` is a stub that echoes "added" without contacting qBittorrent “either implement it or document that the extension targets the Python backend only.

Each backend addition follows full TDD + four-layer coverage + its own code-review GO.

4. Phased implementation plan

Each phase lists **Tasks** → **fine-grained Sub-tasks**, the **traceability IDs** it satisfies (see `_plan/C`), the **adopt/refactor/rewrite/net-new** disposition (`_plan/F`), and the **test types** that gate it (`_plan/D`). Every phase ends with a code-review GO (§11.4.125) before merge. Phases are subagent-driven (§11.4.70), parallelized where file-scopes are disjoint (§11.4.103).

Phase 1 “Foundation & scaffolding (*Disposition: REFACTOR configs / ADOPT types & shared libs*)

Covers: build/config, data model, shared libs, constants, types. Traceability: FND-, ER-, NFR-maintainability. - **T1.1 Scaffold WXT project** “extension/ with `package.json` (SSH repo URL, node ≥20, ESM), `wxt.config.ts` (MV3, minimum_chrome_version 109), `tsconfig.json` (strict, zero any), `.eslintrc/.prettierrc`. **NO .github/**. - Sub: port the npm scripts (dev/build/zip/compile/lint/test) “drop CI-only scripts. - Sub: WXT auto-icons → rasterize the missing icon PNGs from `assets/icon.svg` (net-new). - **T1.2 Manifest/permissions/CSP** “least-privilege per `_plan/E`: storage, alarms, notifications, activeTab, contextMenus; host_permissions localhost-only (:7187 + optional remote); MV3 CSP object form; **no** <all_urls>/tabs/cookies. Re-target all :8080 → :7187. - **T1.3 Types & data model** “ADOPT types/torrent.ts; de-dup AuthMethod/ServerConfig drift; reconcile the SQL schema (9 tables) vs

chrome.storage.JSON â€” **decision:** chrome.storage.local/session is the source of truth for v1 (sql.js deferred unless needed); document the ER model as the logical schema. - **T1.4 Shared libs** â€” ADOPT crypto.ts (sound primitive), errors.ts, events.ts, logger.ts, storage.ts, utils.ts; fix the RateLimitError/ServerError name mismatch; constants port (ports/URLs/timeouts/STORAGE_KEYS). - **Tests:** unit (shared libs, constants, type guards) + the foundation Challenge (artifact-open Â§11.4.38). Vitest configured with UI/entrypoints in coverage.

Phase 2 â€” Core detection & parsing engine (ADOPT parsers/scanners + REFACTOR)

Covers: FR-001..FR-010 (detection/parsing/infoshash/dedup), Dim06/Dim07/Dim08, EDGE-*. - **T2.1 Parsers** â€” ADOPT bencode.ts, magnet.ts; **REFACTOR** torrent-file.ts (wire the dead .torrent parser; SHA-1 infoshash = SHA-1(bencode(info)) via WebCrypto; private-passkey sanitization). - **T2.2 Scanners** â€” ADOPT link-scanner.ts/text-scanner.ts/content/scanner.ts; **REFACTOR** scanner/base.ts (remove Date.now()-salted unstable IDs â†’ stable infoshash-based IDs); **merge the two divergent site-db.ts selector tables** into one; MutationObserver 500ms debounce; Shadow-DOM recursion; iframe all_frames; 16ms frame budget + cooperative yielding. - **T2.3 Dedup** â€” by lowercase infoshash (FR-007). - **Tests:** unit (every bencode type, every magnet param xt/dn/tr/xl/ws/x.pe, hex+base32 infoshash, malformed rejection, infoshash determinism) â€” **remediate** the bencode dead-assert; performance (â‰‰5ms/link on 1000-link page); load (10k-link page, no frame >16ms); stress (Â§11.4.85 boundary inputs); benchmark.

Phase 3 â€” Extension shell (content / background / popup / options) (REFACTOR)

Covers: FR-011..FR-018 (UI, context menu, shortcuts, badge), message protocol, lifecycle/state machines. - **T3.1 Content script** â€” REFACTOR content/index.ts (unify the get-detected message contract); highlight UI; message passing to background. - **T3.2 Background SW** â€” REFACTOR background/index.ts (message router; context menus Ã—4; chrome.commands Ã—3â€”4; chrome.alarms keep-alive ~20s + health 5min; badge/notifications); **remove the empty-passphrase decrypt** (â†’ Phase 7 secure model). Offline-queue state machine (_analysis/03 #09): Idleâ†’Pendingâ†’{Sending|Queued|Retry}â†’{Completed|Failed}, backoff 5sâ†’5min, â‰‰5 retries, max 1000, Dead-Letter. - **T3.3 Popup** â€” REFACTOR popup/popup.ts (detected list, Send, Send-All, status icon); every control â†’ ExtensionConfig field. - **T3.4 Options** â€” REFACTOR options/options.ts (7 tabs: Server/Download/Queue/Notifications/Detection/UI/Security); **remove the fixed "bobalink-extension" passphrase**. - **Tests:** e2e (real extension loaded via persistent-context --load-extension, real id resolved from SW target â€” fixes the test-id bluff); UI/UX (WCAG 2.1 AA, keyboard nav, ARIA); integration (message protocol).

Phase 4 â€” Boba backend integration (NET-NEW + the BE-1/2/3 backend additions)

Covers: FR-019..FR-023 (send, queue, health, auth), Dim01/Dim09 (Dim01 authoritative). - **T4.1 Real Boba client** â€” NET-NEW api/boba-client.ts: POST /api/v1/download (magnet + URL), GET /health, GET /api/v1/auth/status; retry (exp backoff + jitter, 3Ã—, 1sâ†’30s); TokenBucket rate-limit (FR-025); AbortController timeout; auto-discovery of :7187 (+ 7186/7189 fallback probes). - **T4.2 Offline queue auth + persistence** â€” REFACTOR api/queue.ts (inject the real send path; the reference queue never authenticated); durability across SW restart (100%, NFR). - **T4.3 Backend additions (Boba side)** â€” BE-1 CORS, BE-2 /api/v1/download/upload, BE-3 Go stub decision

(Â§3 above). TDD on the Python/Go backend. - **Tests:** integration against live 7187 (require_backend(7187) â†’ SKIP-with-reason if down, never fail-open Â§11.4.3/Â§11.4.68); security (no creds sent to qBittorrent from the extension); chaos (backend drops mid-send â†’ queue recovers); the end-to-end Challenge (detectâ†’sendâ†’**torrent appears in qBittorrent**).

Phase 5 â€” Tab-group batch (*NET-NEW â€” does not exist in the reference*)

Covers: FR-024, Dim05, sequence #05. - **T5.1** chrome.tabGroups.query â†’ chrome.tabs.query{groupId} â†’ chrome.scripting.executeScript per tab â†’ scan â†’ flatten + dedup â†’ batch send (urls=magnet1\nmagnet2â€¦). - **T5.2** Context-menu â€œSend tab groupâ€” + popup â€œSend All in groupâ€” collapsed-group handling; Firefox-incompatible â†’ graceful degrade. - **Tests:** e2e (real multi-tab group), integration (batch dedup), UX (progress notification), Challenge (tab-group batch â†’ N torrents in qBittorrent).

Phase 6 â€” UI/UX, i18n, accessibility, themes (*REFACTOR/ADOPT*)

Covers: FR-016/FR-017, Dim10, NFR-ally. - **T6.1** i18n (_locales â€” 8 locales; default_locale); **T6.2** dark/light via prefers-color-scheme; **T6.3** WCAG 2.1 AA audit (contrast, focus, ARIA, keyboard); **T6.4** notification templates (no Base64 icons; Mac limits); badge â‰¥4 chars. - **Tests:** UX (ally automated + heuristic), UI (every control state), i18n (locale switch renders).

Phase 7 â€” Security & credentials (*NET-NEW key source + REFACTOR call-sites; crypto module ADOPTED*)

Covers: SEC-*, Dim11, diagram #10, Â§11.4.10. Full design: _plan/E. - **T7.1 Credential model** â€” delegate-by-default (extension holds **no decryptable secret** in the localhost-proxy deployment); residual remote-Boba token â†’ user-supplied session passphrase (PBKDF2, chrome.storage.session only, never persisted). Remove both broken call-sites. - **T7.2 Controls** â€” least-privilege manifest, MV3 CSP, HTTPS-off-localhost + cert verify, magnet/dn XSS sanitization, passkeyâ†’***PASSKEY*** log redaction, no-cleartext-credential-logging, sender.id/sender.url validation, GDPR â€œClear All Data.â€” - **Tests (anti-bluff, each RED-on-broken first Â§11.4.115, permanent guards Â§11.4.135):** T2 wrong-key fails (the "boba-link-extension" + "" decrypts both THROW), T3 no embedded key (grep src + bundle â†’ zero hits), T4 no secret in logs, T1 ciphertext-at-rest, T8 delegation proof (no qBt/tracker creds in extension state); security/pen-test suite; Â§11.4.10.A pre-store leak audit + pre-commit grep gate.

Phase 8 â€” Testing to 100% (all types) + Challenges + HelixQA (*NET-NEW*)

Covers: the 13 mandated test types, every feature. Full matrix: _plan/D. - **T8.1 Remediate the 10 reference bluffs** (B1â€”B10): inline-cryptoâ†’import real module; Jestâ†’Vitest; expect(true) no-opsâ†’real assertions; non-functional e2eâ†’real-extension fixture + global-setup/teardown; coverage includes UI/entrypoints. - **T8.2 Fill missing test types** â€” integration, security, load (10k links + API flood vs rate-limiter), scaling, chaos (backend death, storage-quota, SW termination), stress (Â§11.4.85 â‰¥100 iters/â‰¥10 concurrent, p50/p95/p99), performance + benchmark (NFR thresholds), UI, UX. - **T8.3 Challenges** â€” 8 challenges/scripts/boba_ext_*.sh driving the **real built extension** end-to-end (detectâ†’sendâ†’qBittorrent row; offline-queue durability; cred-at-rest; auto-discover; tab-group batch; artifact-open Â§11.4.38; cred-leak grep). No false success. - **T8.4 HelixQA banks** â€” submodules/helixqa/banks/boba-boba-link.yaml (http: style, backend endpoints, CI-gated) + boba-boba-link-ui.yaml (Playwright, extension UI). Resolve the

helixqa build-blocker (add the 6 sibling submodules **or** consume bank-YAML only). Autonomous QA session, re-runnable N/A — (Â§11.4.98/Â§11.4.50), live JSONL verdict stream (Â§11.4.116). - **T8.5** Coverage ledger â†’ 100% feature — test-type; ratchet thresholds; every green proves a user-observable outcome.

Phase 9 â€” Build, packaging & distribution (manual â€” NO CI/CD)

Covers: BLD-*, Dim12, diagram #11. - **T9.1** `extension/ci-ext.sh` (manual gate: `lint â†’ format:check â†’ tsc --noEmit â†’ vitest â€”“coverage â†’ playwright â†’ wxt build â†’ zip + manifest-validate + bundle-size â‰‰350KB`), shelled from `./ci.sh`. **NO GitHub Actions / GitLab pipelines.** - **T9.2** Per-store packaging (Chrome/Firefox-gecko.id/Opera/Edge/Yandex) â€” **manual** zips; document submission steps; no auto-submit, no store secrets in git. - **T9.3** User + developer + installation + API docs (port the 5 reference guides, corrected for the real 7187 contract), with Â§11.4.65 html/pdf siblings.

5. Test/Challenge/HelixQA coverage strategy (summary â€” full in `_plan/D`)

- **Runner:** Vitest + Playwright (matches Boba frontend/; one coverage tool @`vitest/coverage-v8`).
- **E2E:** `wxt build â†’ launchPersistentContext + --load-extension â†’ real id from SW target`.
- **Live integration:** `require_backend(7187)` SKIP-with-reason if down (never fail-open).
- **Coverage:** UI/entrypoints **in scope**; ratchet â‰‰90%â†’100; anti-bluff is the real gate.
- **Per-feature — per-test-type matrix:** 14 types — 13 modules; N/A cells justified (DDoS N/A for a client parser, but Load IS in-scope for 10k-link pages + rate-limiter flood).

6. Adopt/Refactor/Rewrite map (summary â€” full in `_plan/F`)

~80% reusable. ADOPT: parsers (bencode/magnet), scanners, shared libs incl.Â the crypto **module** (sound), types, i18n. REFACTOR: api targets (8080â†’7187), torrent-file wiring, scanner IDs/site-db, the two credential call-sites. REWRITE: the test suiteâ€™s bluffs + configs. NET-NEW: real Boba client, tab-groups, secure key source, the 13 missing test types, icon PNGs. DROP: both `.github/workflows/*.yml`.

7. Open operator decisions

1. **Subproject location** â€” `extension/` at repo root (proposed). Object if you want it elsewhere.
2. **Backend additions** â€” OK to add BE-1 (CORS) + BE-2 (`/api/v1/download/upload`) to the Python merge-service? (Required for `.torrent-bytes` upload + `extension-origin` calls.)
3. **HelixQA build-blocker** â€” add the 6 missing sibling submodules (`doc_processor`, `llm_orchestrator`, `llm_provider`, `llms_verifier`, `security`, `vision_engine`) so `bin/helixqa` builds in-repo, **or** consume bank-YAML only and run `helixqa` from a standalone checkout?

4. **Session scope** – proceed to Phase 1 scaffolding now (autonomous), or hold for your review of this plan first?
5. **Product name** BobaLink – keep?

8. Traceability & provenance

- Full requirements matrix (245 items, 240 v1 / 5 v2): `_plan/C-traceability-matrix.md` – every FR/NFR/Dim/ER/edge-case placed; none untraceable.
- Source extractions (every file read, manifests prove coverage): `_analysis/01-06.md`.
- Planning evidence: `_plan/A` (backend contract), `B` (HelixQA), `C` (traceability), `D` (tests), `E` (security), `F` (adopt/rewrite).
- v2-deferred (documented seams, not dropped): multi-client gateway, `search_cache`, Boba search/SSE, side-panel, qBt API-key auth.